

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$H \ell$
									$ \begin{aligned} & + C_{H2} H_{a_1} H_{a_1}^\dagger \\ & - H_{a_1} \mu^2 H_{a_1}^\dagger \\ & + \frac{3h H_{a_1} H_{a_2} \lambda_{H1}^2 H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ & + \frac{3}{2} \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_{S_1}^2} \right) H_{a_1} H_{a_2} \lambda_{H1}^2 H_{a_1}^\dagger H_{a_2}^\dagger \\ & + \frac{3h H_{a_1} H_{a_2} \lambda_{H2}^2 H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ & + \frac{3}{2} \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_{S_2}^2} \right) H_{a_1} H_{a_2} \lambda_{H2}^2 H_{a_1}^\dagger H_{a_2}^\dagger \\ & + \frac{\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & + \frac{\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \\ & + \frac{\hbar \log \left(\frac{\mu_{\text{bar}2}}{M_{S_1}^2} \right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \\ & - \frac{\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \\ & - \frac{\hbar \log \left(\frac{\mu_{\text{bar}2}}{M_{S_2}^2} \right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \end{aligned} $	$ \begin{aligned} & - \frac{5\hbar \left(W_{\mu_1 \mu_2}^{A_1} \right)^2}{6\epsilon} \\ & - \frac{1}{2} \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_F^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2 \\ & - \frac{1}{6} \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_{S_1}^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2 \\ & - \frac{1}{6} \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_{S_2}^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2 \end{aligned} $	$ \begin{aligned} & i\hbar \log \left(\frac{\mu_F^2}{M_F^2} \right) \frac{\overline{W_{1,1,1} W_{2,12,1}^*} \ell_{a_4,12}^T C P_L \ell_{a_2,11} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T1} \ell_{a_1,12} \ell_{a_3,12} M_F^2}{2(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_1}^2)(M_{S_1}^2 - M_{S_1}^2)} \\ & + i\hbar \log \left(\frac{\mu_F^2}{M_{S_1}^2} \right) \frac{\overline{W_{1,1,1} W_{2,12,1}^*} \ell_{a_4,12}^T C P_L \ell_{a_2,11} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T1} \ell_{a_1,12} \ell_{a_3,12} M_F^2}{2(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_1}^2)(M_{S_1}^2 - M_{S_1}^2)} \\ & + i\hbar \log \left(\frac{\mu_F^2}{M_{S_1}^2} \right) \frac{\overline{W_{1,1,1} W_{2,12,1}^*} C P_R \ell_{a_4,12}^T M_{S_1}^2 W_{1,1,1} W_{2,12,1} \ell_{a_1,12} \ell_{a_3,12} H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{2(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_1}^2)(M_{S_1}^2 - M_{S_1}^2)} \\ & + i\hbar \log \left(\frac{\mu_F^2}{M_{S_1}^2} \right) \frac{\overline{W_{1,1,1} W_{2,12,1}^*} C P_R \ell_{a_4,12}^T M_{S_1}^2 W_{1,1,1} W_{2,12,1} \ell_{a_1,12} \ell_{a_3,12} H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{2(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_1}^2)(M_{S_1}^2 - M_{S_1}^2)} \\ & + i\hbar \log \left(\frac{\mu_F^2}{M_{S_1}^2} \right) \frac{\overline{W_{1,1,1} W_{2,12,1}^*} \ell_{a_4,12}^T C P_L \ell_{a_2,11} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T1} \ell_{a_1,12} \ell_{a_3,12} M_F}{2(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_1}^2)(M_{S_1}^2 - M_{S_1}^2)} \\ & + i\hbar \log \left(\frac{\mu_F^2}{M_{S_1}^2} \right) \frac{\overline{W_{1,1,1} W_{2,12,1}^*} \ell_{a_4,12}^T C P_L \ell_{a_2,11} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T1} \ell_{a_1,12} \ell_{a_3,12} M_F}{2(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_1}^2)(M_{S_1}^2 - M_{S_1}^2)} \\ & + i\hbar \log \left(\frac{\mu_F^2}{M_{S_1}^2} \right) \frac{\overline{W_{1,1,1} W_{2,12,1}^*} C P_R \ell_{a_4,12}^T M_{S_1}^2 W_{1,1,1} W_{2,12,1} \ell_{a_1,12} \ell_{a_3,12} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{2(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_1}^2)(M_{S_1}^2 - M_{S_1}^2)} \\ & + i\hbar \log \left(\frac{\mu_F^2}{M_{S_1}^2} \right) \frac{\overline{W_{1,1,1} W_{2,12,1}^*} C P_R \ell_{a_4,12}^T M_{S_1}^2 W_{1,1,1} W_{2,12,1} \ell_{a_1,12} \ell_{a_3,12} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{2(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_1}^2)(M_{S_1}^2 - M_{S_1}^2)} \end{aligned} $
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$ \begin{aligned} & - \frac{3\hbar B_{\mu_1 \mu_2}^2}{4\epsilon} \\ & - \frac{1}{2} \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_F^2} \right) B_{\mu_1 \mu_2}^2 \\ & - \frac{1}{4} \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_{S_2}^2} \right) B_{\mu_1 \mu_2}^2 \end{aligned} $			

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$$\begin{aligned}
& \frac{3h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{16 (M_F^2 - M_{S_1}^2)^2} \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^4 \\
& - \frac{h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{4 (M_F^2 - M_{S_2}^2)^2} \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^4 \\
& - \frac{3h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{16 (M_F^2 - M_{S_1}^2)^2} \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^4 \\
& - \frac{h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{4 (M_F^2 - M_{S_2}^2)^2} \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^4 \\
& + \frac{3h y_{1,i_3,r_1} \ell_{a_1,i_2} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S_1}^2)} \\
& + \frac{3h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{8 (M_F^2 - M_{S_1}^2)} \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2 \\
& + \frac{3h \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{16 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{3h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{8 (M_F^2 - M_{S_1}^2)^2} \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2 \\
& + \frac{h \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{2 (M_F^2 - M_{S_2}^2)} \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2 \\
& + \frac{h \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{2 (M_F^2 - M_{S_2}^2)^2} \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2 \\
& + \frac{3h \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{8 (M_F^2 - M_{S_1}^2)} \\
& + \frac{3h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{8 (M_F^2 - M_{S_1}^2)} \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2 \\
& + \frac{3h \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{16 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{3h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{8 (M_F^2 - M_{S_1}^2)^2} \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2 \\
& + \frac{h \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{2 (M_F^2 - M_{S_2}^2)} \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2 \\
& + \frac{h \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{4 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{2 (M_F^2 - M_{S_2}^2)^2} \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2 \\
& + \frac{3h \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{16e} \\
& - \frac{3}{32} h \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{3h \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{8 (M_F^2 - M_{S_1}^2)} \\
& - \frac{3h \log \left(\frac{\mu h \nu_2^2}{M_{S_1}^2} \right)}{8 (M_F^2 - M_{S_1}^2)} \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{3h \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{16 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{h \overline{y_{2,i_2,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4e} \\
& - \frac{1}{8} h \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{h \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{h \log \left(\frac{\mu h \nu_2^2}{M_{S_2}^2} \right)}{2 (M_F^2 - M_{S_2}^2)} \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{h \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{h \log \left(\frac{\mu h \nu_2^2}{M_{S_2}^2} \right)}{4 (M_F^2 - M_{S_2}^2)^2} \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1} \\
& + \frac{3h \overline{y_{1,i_2,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} y_{1,i_3,r_1} H_{a_1}^\dagger}{16e} \\
& - \frac{3}{32} h \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} y_{1,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{3h \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{8 (M_F^2 - M_{S_1}^2)} \\
& + \frac{3h \log \left(\frac{\mu h \nu_2^2}{M_{S_1}^2} \right)}{8 (M_F^2 - M_{S_1}^2)} \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{3h \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} M_{S_1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{16 (M_F^2 - M_{S_1}^2)^2}
\end{aligned}$$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	e
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